Can a model find a water leak faster than guessing?

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25.9%

of the water taken in by Chișinău was accepted as lost in the network in the 2026 tariff calculation

1,037 m³

a day from one 12 L/s leak: the daily drinking water of about 7,975 people (130 L each)

Question and hypotheses

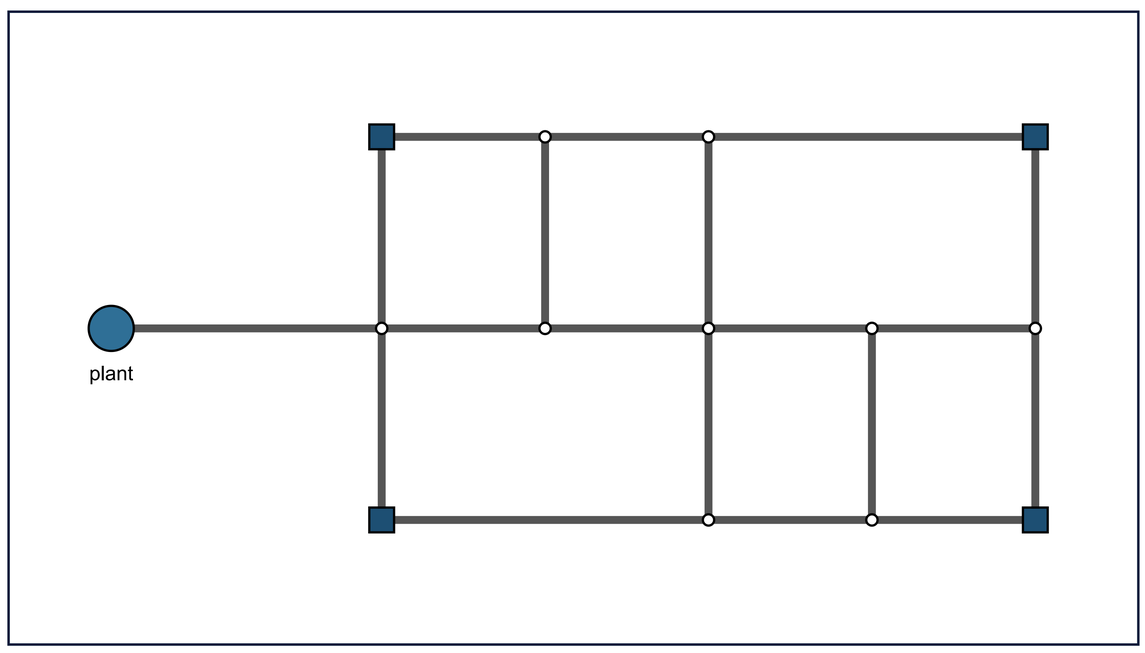
A leak opens and only four pressure sensors can see it. How many wasted digs (“dry holes”) does a crew need, and does a hydraulic model of the network help? Written down **before any method was run** (Git history shows the order).

> **H1** Model fitting needs at least 50% fewer dry holes than the naive rule. **PARTIALLY SUPPORTED**
> Observed 13.6% fewer (95% interval 10.0% to 17.2%).

> **H2** The cheap fingerprint method is within 5 points of model fitting and 10 times faster. **PARTIALLY SUPPORTED**
> 8,035 times faster, but the gap is 3.5 points (upper bound 6.6).

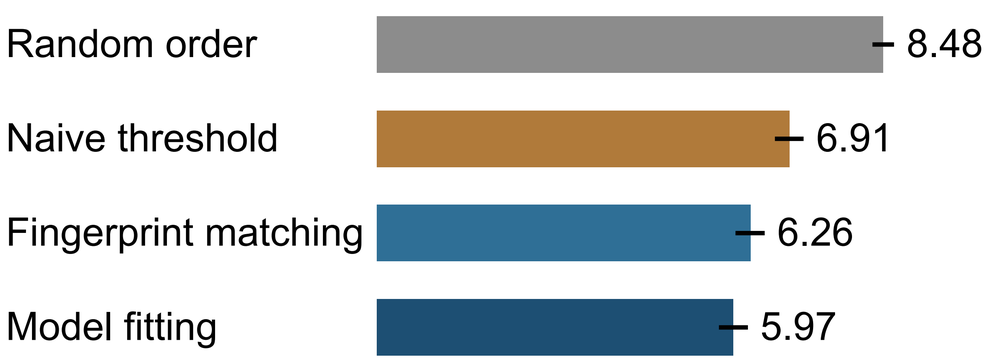
The targets were missed and were not changed afterwards.

Method



A simulated city: 13.4 km of pipe, 14 junctions, 6 loops, 4 pressure sensors (squares). Hazen–Williams hydraulics, Global Gradient Algorithm. 1,000 random leaks of 5–25 L/s; the methods do not know the real demand (±10% per junction); sensor error 0.10 m. Four methods choose where to dig, with the same crew rule.

Results

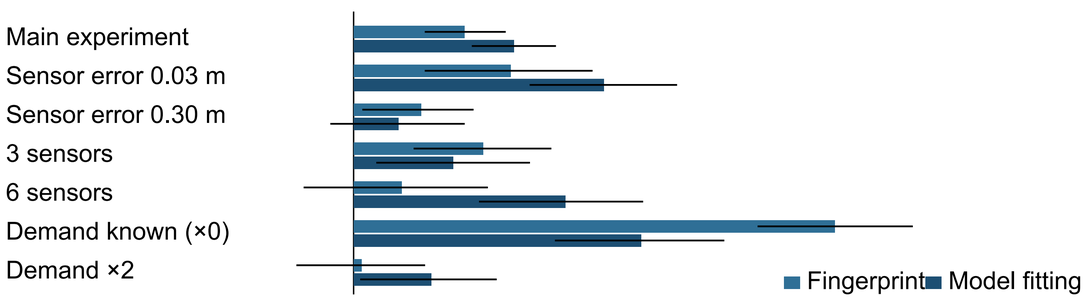


Mean dry holes before the leak is found (95% interval); fewer is better; 10 = not found.

Method	Dry holes	Within 3 digs	Never found
Random order	8.48	8.7%	72.8%
Naive threshold	6.91	19.7%	53.7%
Fingerprint matching	6.26	25.8%	44.2%
Model fitting	5.97	29.3%	38.8%

Model fitting never found **38.8%** of the leaks in 10 digs. Fingerprints take 0.03 ms; model fitting takes 243 ms.

What decides the result



Dry holes saved against the naive rule. With demand known exactly, fingerprints save **41%**; with twice the uncertainty, **1%**. **Knowing the demand matters more than the algorithm.**

Money, in lei and euro

	Today	Demand known
One dry hole	€980 (19,781 lei)	
Payback, model fitting	66 months	31 months
Payback, fingerprints	138 months	16 months

Tariff 14.03 lei/m³ (ANRE, 4 Aug 2026); €1 = 20.18 lei (BNM, 5 Nov 2026). A €12,000 system, 4 large leaks a year; break-even 4.2 leaks a year for a five-year payback. **The dig cost, leaks per year and system cost are assumptions.**

Limits and honesty

- A simulation of a drawn network: no real leaks, sensors or Moldovan data.
- The model is assumed exact; one leak; steady flow.
- Low-cost transmitters (±0.5–1% of full scale = 0.5–1 m on 10 bar) are noisier than the 0.10 m assumed.
- Next: a real district, a flow meter at the plant, several leaks.

QR: live site

QR: source code

drop-wise-s8pt.vercel.app
github.com/nicolepuzur/DropWise

AI cited: Claude Code (Anthropic), model claude-sonnet-5-5 — every prompt and what was checked: docs/AI_USAGE.md.
Code MIT-licensed.